

AspeQt-2k26

High-Performance Peripheral Emulator for Atari 8-Bit Computers

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Introduction

AspeQt-2k26 is a modernized version of the classic AspeQt emulator, updated to use the Qt6 framework. It transforms your modern PC into a powerful peripheral server for Atari 8-bit computers (400/800/XL/XE).

Using a standard SIO2PC cable (USB or Serial), AspeQt can emulate:

- **Disk Drives (D1:-D8:)** - Supports ATR, XFD, PRO images, and direct Folder Mounting.
- **TNFS Client** - Mount and stream disk images directly from the internet (FujiNet protocol).
- **Printers (P:)** - Captures printed output to text or graphical view.
- **Clipboard Bridge (Y:)** - Bi-directional copy/paste between PC and Atari.
- **Web Pipe (W:)** - Thin-client gateway for Python scripts and AI.
- **Cassette (C:)** - Load CAS files via SIO audio.

High-Speed SIO

AspeQt supports "divisor 0" high-speed SIO (approx. 126kbps) with custom OS ROMs (like QMEG or Hiassoft) and standard high-speed protocols (UltraSpeed, Happy, etc.).

Note: Reliable high-speed operation depends on your SIO2PC hardware. FTDI-based USB cables generally offer the best performance.

Disk & Folder Usage

Mounting Disk Images

Drag and drop `.atr`, `.xfd`, or `.pro` files directly into the drive slots (D1-D8). Double-clicking a slot opens the file browser.

Virtual DOS (Folder Images)

You can mount a local PC directory as a writable disk. To boot from a folder, right-click the slot and select **Mount Folder**.

Supported DOS structures for folder mounting:

- Atari DOS 2.0 / 2.5
- MyDOS 4.5
- SpartaDOS (Limited)

Running Executables

Drag `.xex` or `.com` files directly to a slot. AspeQt uses a high-speed internal loader to execute these files instantly, bypassing the need for a DOS.

TNFS Client

AspeQt-2k26 includes native support for TNFS (The Network File System), allowing you to mount and stream disk images directly from the internet using the FujiNet protocol.

Connecting to a Server

Access the TNFS Client via the **Tools** menu. You can add standard TNFS servers (such as `fujinet.pl` or `atari-apps.irata.online`).

Usage

Once connected, the remote file system appears in the AspeQt file browser. You can navigate folders and drag remote `.ATR` or `.XEX` files directly into the drive slots (D1-D8). AspeQt will stream the data on-demand, allowing you to boot software immediately without manually downloading files.

Printer Emulation (P:)

Output sent to `P:` is captured by AspeQt. View the output via **Tools > View Printer Output**. The viewer automatically handles ATASCII-to-ASCII translation.

The Y: Device (Clipboard Bridge)

The **Y: Device** is a custom CIO handler that bridges the gap between your Atari 8-bit and your modern PC's system clipboard. It allows you to "Paste" code from your PC directly into Atari BASIC, or "Copy" listings from the Atari back to your PC.

1. Installation

The Y: device is not built into the Atari OS; it must be loaded into memory.

1. Load the **handler.xex** file provided with AspeQt (drag it to D1: and boot, or load it from DOS).
2. The handler installs itself at memory address **\$4000**.
3. Once loaded, the device **Y:** becomes available to the system.

2. Importing Code (PC → Atari)

This feature allows you to write BASIC code in a modern editor (like VS Code or Notepad++) and run it on the Atari.

- **On PC:** Select your text/code and **Copy** (Ctrl+C).
- **On Atari:** Type **NEW**, then type **ENTER "Y: "**

The Atari will read the text from your clipboard as if it were typing it in. AspeQt automatically converts modern Line Feeds (**\n**) into ATASCII EOL (**\$9B**) characters.

3. Exporting Code (Atari → PC)

This feature allows you to save your work from the Atari back to your PC clipboard.

- **On Atari:** Type **LIST "Y: "**
- **On PC:** Open your text editor and **Paste** (Ctrl+V).

Note: Opening "Y:" for writing (Mode 8) automatically **clears** the PC clipboard before appending the new data.

4. Advanced Usage (BASIC Example)

The Y: device functions like any standard IOCB device. You can write your own software to utilize the clipboard bridge. Below is a BASIC example demonstrating both reading and writing.

```
10 REM === WRITE TO CLIPBOARD ===  
20 OPEN #1,8,0,"Y:"
```

```
30 ? "Writing to PC Clipboard..."
40 PRINT #1;"Hello from the 8-bit world!"
50 PRINT #1;"This text was sent via AspeQt."
60 CLOSE #1
70 REM === READ FROM CLIPBOARD ===
80 ? "Copy something on PC, then press Return."
90 DIM A$(1):INPUT A$
100 TRAP 150 :REM Catch EOF Error
110 OPEN #1,4,0,"Y:"
120 GET #1,B
130 ? CHR$(B);
140 GOTO 120
150 CLOSE #1
160 ? "Done."
```

The W: Device (Web Pipe)

The **W:** device transforms the Atari 8-bit into a "Thin Client" capable of interacting with modern web services. Instead of implementing complex network stacks (TCP/IP, SSL) on the Atari, the **W:** device acts as a transparent proxy. It forwards standard BASIC text commands to a backend script running on your PC via HTTP.

How It Works

The device operates using standard Atari I/O commands:

- **Write Mode (HTTP POST):** Opening W: for write allows you to buffer data. When you CLOSE the channel, AspeQt sends the buffered data as an HTTP POST request to the specified URL.
- **Read Mode (HTTP GET):** Opening W: for read immediately performs an HTTP GET request to the URL. Subsequent INPUT commands retrieve the server's response.

Running the Handler

To enable the W: and Y: devices, you must first load the handler into memory.

- **Manual Loading:** From the DOS menu, select option **L** (Binary Load) and type `HANDLER.XEX`.
- **Automatic Loading:** Rename `HANDLER.XEX` to `AUTORUN.SYS` on your boot disk. This ensures the network drivers load automatically every time you turn on the computer.

Code Example: The "Hello World" Loop

This example demonstrates sending data to a Python script and reading the response.

1. Atari BASIC (The Frontend)

```

10 DIM U$(40), MSG$(40), REPLY$(40)
20 U$="W:http://192.168.1.5:5000/atari"
30 MSG$="HELLO FROM 1979"

40 REM --- STEP 1: SEND DATA (POST) ---
50 OPEN #1,8,0,U$      : REM Open for Write
60 PRINT #1;MSG$       : REM Send the message
70 CLOSE #1           : REM Trigger the Network Request

80 REM --- STEP 2: READ RESPONSE (GET) ---
90 OPEN #1,4,0,U$      : REM Open for Read
100 INPUT #1,REPLY$    : REM Read text from PC
110 PRINT "SERVER SAID: ";REPLY$
120 CLOSE #1

```

2. Python/Flask (The Backend)

```

from flask import Flask, request

app = Flask(__name__)

@app.route('/atari', methods=['GET', 'POST'])
def handle_atari():
    if request.method == 'POST':
        # Receive text from Atari
        data = request.data.decode('utf-8')
        print(f"Atari sent: {data}")
        return "OK"
    else:
        # Send text to Atari
        return "MESSAGE RECEIVED LOUD AND CLEAR"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)

```

Native FTP Support (GET & PUT)

AspeQt-2k26 includes a built-in bridge to the **FTP protocol**. This allows your Atari to download files, read directory listings, and upload data directly to FTP servers without needing a PC-side Python script.

Prerequisite: This feature requires the **curl** utility to be installed on your Host PC (Windows/Linux/Mac) and

available in the system PATH.

1. Reading Files & Directory Listings (GET)

To read a file or view a directory, simply open the W: device with an `ftp://` URL in Read mode (4).

- **Files:** Text files are streamed to the Atari. Unix Line Feeds (`\n`) are automatically converted to Atari EOLs (`$9B`).
- **Directories:** If the URL ends in a slash (`/`), AspeQt returns a directory listing.

```
10 DIM U$(60), L$(120)
20 REM --- PUBLIC FTP EXAMPLE ---
30 U$="W:ftp://ftp.funet.fi/pub/atari/README"
40 OPEN #1,4,0,U$
50 TRAP 90
60 INPUT #1,L$
70 PRINT L$
80 GOTO 60
90 CLOSE #1
100 PRINT "--- EOF ---"
```

2. Uploading Files (PUT)

To upload data, open the W: device in Write mode (8). You must provide valid credentials in the URL if the server requires them.

```
10 DIM U$(80)
20 REM --- UPLOAD WITH CREDENTIALS ---
30 U$="W:ftp://user:password@192.168.1.50/upload.txt"
40 OPEN #1,8,0,U$
50 PRINT #1;"This text is being uploaded from an Atari 8-bit!"
60 CLOSE #1 : REM Upload starts immediately on CLOSE
```

Application Examples Library

This library contains a collection of "Hybrid Computing" examples that demonstrate the power of the W: Device. These examples offload heavy processing (AI, Web Scraping, SSL, Speech Synthesis) to a modern PC (Host) while using the Atari 8-bit as the frontend terminal.

General Prerequisites

For all examples below, you must have:

- **Python 3.x** installed on your Host PC.
- **Flask** installed (`pip install flask`).
- The **PC's IP Address** configured in the BASIC .LST files.

1. Gemini AI Chatbot

Turns the Atari into a client for Google's Gemini Large Language Model. You can type natural language questions ("Who is Captain Kirk?") and receive intelligent, summarized answers on the Atari screen.

- **BASIC File:** GEMINI_AI.LST
- **Python Script:** gemini_ai.py
- **Dependencies:** `pip install google-generativeai`

2. Basic Graphical Weather Station

Fetches live weather data from the Open-Meteo API for a specific location and displays it on the Atari using GRAPHICS 5 (4-color mode).

- **BASIC File:** WEATHER.LST
- **Python Script:** weather.py
- **Dependencies:** `pip install requests`

3. Cloud Notepad

Demonstrates Reading and Writing to the PC's hard drive. It acts as a "Cloud Drive" for text.

- **BASIC File:** CLOUD_NOTE.LST
- **Python Script:** cloud_note.py

4. Text-to-Speech Synthesizer

The "Talking Atari." Any text typed on the Atari is sent to the PC, where it is read aloud using the Linux espeak engine.

- **BASIC File:** TTS_SERVER.LST
- **Python Script:** tts_server.py
- **Dependencies:** `sudo apt install espeak` (Linux)

5. Web News Scraper

Connects to a modern news website, scrapes the HTML, strips out ads/javascript, and delivers a plain-text headline summary to the Atari.

- **BASIC File:** SCRAPER.LST
- **Python Script:** scraper.py
- **Dependencies:** `pip install requests beautifulsoup4`

6. Remote Execution (Linux & Windows)

Allows the Atari to execute shell commands or launch GUI applications on the Host PC.

- **BASIC Files:** EXEC_LINUX.LST / EXEC_WIN.LST
- **Python Scripts:** exec_linux.py / exec_win.py

7. File Upload / Telemetry

A utility to upload raw data or text files from the Atari to the PC.

- **BASIC File:** FILE_UPLOAD.LST
- **Python Script:** file_upload.py

Quick Dependency Install

```
# Linux/Mac
sudo apt install espeak
pip install flask requests beautifulsoup4 google-generativeai psutil

# Windows
pip install flask requests beautifulsoup4 google-generativeai psutil
```